Food Policy 121 (2023) 102545

Contents lists available at ScienceDirect

Food Policy

journal homepage: www.elsevier.com/locate/foodpol

Economic impact assessment of public incentives to support farm-to-school food purchases

Shayna M. Krasnoff [a], Todd M. Schmit [a,*], Cheryl B. Bilinski [b]

[a] Charles H. Dyson School of Applied Economics and Management, Cornell University, Warren Hall, Ithaca, NY 14853, United States
[b] Local Food Systems Specialist, Cornell Cooperative Extension, Ithaca, NY 14853, United States

ARTICLE INFO

Keywords:
Farm-to-school
Local foods
Economic impact
Food policy

ABSTRACT

Farm-to-school projects have been widely supported by policy makers with funding provided at state and federal levels. Still, many of the outcomes of this inflow of policy and funding remain unclear, often due to insufficient data to examine them. In 2018, New York State (USA) announced the 30% NY Initiative that substantially increases school lunch reimbursements if school districts purchase at least 30% of their ingredients as New York food products. With detailed food purchasing data from the second largest school district in the state and the largest to qualify for enhanced reimbursement, we estimate the gross and net economic impacts of the policy through a customized input–output model. We observe clear shifts in food spending categories that suggest changes in what and where foods were purchased. Results demonstrate net positive value added impacts of the policy even when a negative impact is applied to account for the cost of the policy to taxpayers. For every dollar in gross domestic product lost in the state to support the program, $1.06 of gross domestic product is expected to be added. However, the results are only true to the extent that the increase in local food spending is commensurate with an expansion of the related farm and food product industries to meet that demand. Specifically, at least 67% of the growth in local food spending must contribute to new aggregate demand for the related food product industries, rather than reallocation from other local marketing channels.

1. Introduction

The Farm-to-School (FTS) movement in the United States can be traced back to the late 1990s, partly as a response to health concerns over increasing amounts of processed foods served to students, but also to growing interest in local food policies that support multiple objectives: rural economic development, farm viability, food access and equity, and sustainable food systems, among others. Recent literature discussing local food policies increasingly point to public institutional procurement as a key component of local food systems growth (e.g., Martinez, 2016; DeSchutter et al., 2020; Kovacs, et al., 2020; Cohen and Ilieva, 2021; Clark, et al., 2021; Parsons and Barling, 2022). Indeed, the United States Department of Agriculture (USDA) reports that U.S. farms sales direct to institutions and intermediated markets (including educational institutions) increased almost 23% between 2015 and 2020, with significant variation across space (USDA 2022c). These sales increased over 75% for the Northeast region (USDA, 2022c).[1]

FTS programming has grown significantly in the United States, with the recent Farm-to-School Census reporting over 65% of School Food Authorities (SFAs) participated in FTS activities in the 2018/19 school year (SY) (USDA, 2021a). FTS activities are also supported by USDA's Food and Nutrition Service (FNS), with activities ranging from the serving of locally produced foods in cafeterias to food exposure activities and agricultural education experiences (USDA, 2023). New York State's (NYS) FTS program supports a multiple objective approach and is coordinated with the New York State Department of Agriculture and Markets (NYSDAM) to provide SFAs financial and technical assistance that builds connections between local farms and food producers, prepares children to learn through adequate nutrition, improves health and well-being, strengthens the local economy, and builds healthy communities (NYSDAM, 2022). Over 77% of the state's SFAs participated in FTS programs in the 2018/19 SY (USDA, 2021b).

In 2018, FTS programming in NYS expanded through the state sponsored "No Student Goes Hungry Program." The Program aspires to

* Corresponding author at: Cornell University, 350A Warren Hall, Ithaca, NY 14853-7801, United States.
E-mail addresses: smk385@cornell.edu (S.M. Krasnoff), tms1@cornell.edu (T.M. Schmit), cbt32@cornell.edu (C.B. Bilinski).

[1] The Northeast region (Region 2) includes the states of Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, and Vermont.

https://doi.org/10.1016/j.foodpol.2023.102545
Received 12 August 2022; Received in revised form 6 October 2023; Accepted 14 October 2023
Available online 21 October 2023

combat food insecurity and hunger by providing students with locally grown, nutritious meals, and supporting an improved learning experience for children of all ages (Khalife, 2018). Included in the Program is the "30% NYS Initiative" (Initiative) that provides SFAs with increased State reimbursement for the purchase of New York Food Products (NYFP). The Initiative defines a NYFP as a food item that is grown, harvested, or produced in NYS; or a food item processed inside or outside NYS comprising at least 51% agricultural raw materials grown, harvested, or produced in NYS by weight or volume (Bilinski et al., 2022). SFAs that spend at least 30% of total food costs for their lunch program on NYFPs receive an increased state school lunch reimbursement from 5.9 to 25 cents per meal. To qualify for the enhanced reimbursement, the SFA must participate in the National School Lunch Program (NSLP) that provides free and reduced-price lunch for qualifying students and track the origins of the foods they are serving specifically for lunch.

Although the intended benefits of the program fall in line with the missions of increasing access to healthy foods and strengthening local economies, the actual economic impacts of the policy remain unclear. We develop a customized input–output (IO) model for NYS to estimate the gross and net economic impacts of the Initiative by analyzing actual changes in spending due to participation in the program. Our quasi-experimental approach analyzes the Buffalo City School District's (BCSD) food spending patterns for the SY immediately before implementation of the Initiative (2017/18) and for the first year the Initiative was in effect (2018/19). To accomplish our objective, we utilize detailed school food procurement data distinguished by source of origin.

While the analysis of only one SFA in the state may appear constraining to a robust evaluation of the Initiative, BCSD was one of only seven SFAs to qualify for the Initiative in 2018/19 and, by far, the largest. BCSD lunch expenditures represented 90% of all qualifying SFA food costs for lunch and 90% of total NYFP expenditures (Bilinski et al., 2022). Since the economic benefits and costs of the Initiative relate specifically to the levels of local food spending and state reimbursement, respectively, an overall evaluation of the Initiative is highly dependent on its largest contributors. In short, expenditures drive economic impact and BCSD had an overwhelming share of them.

We continue with a literature summary evaluating public procurement policies of local foods and how our alternative approach contributes importantly to that body of work. We then describe the Initiative in more detail and the data collection and processing efforts necessary for the analysis. The methods and empirical results follow, and we conclude with policy implications of our results and directions for future research.

2. Evaluating food policies on public procurement

Evaluations of food polices commonly investigate impacts to particular stakeholders or industries (e.g., to schools, households, children, or farms) or recommend policy amendments to better address particular goals (e.g., improving diet quality and nutritional outcomes, or food access and sustainability issues). Accordingly, approaches in the literature often investigate associations in policy implementation to particular backward- or forward-linked supply chain effects; e.g., general policies supporting local food system participation and the impact to farm viability and net returns (backward) (e.g., Aubry & Kebir, 2013; Bauman & Jablonski, 2018; Brekken et al. 2019; Schmit et al., 2021) or to food access and nutritional outcomes (forward) (DeSchutter et al., 2020; Cohen and Ilieva, 2021). Early FTS research focus on the social impacts to stakeholders involved and the gains in knowledge obtained (Joshi et al., 2008), while others assess their efficacy as nutritional tools (Graham & Zidenberg-Cherr, 2005; Ozer, 2007).

In the context of public procurement policies for local foods, a range of empirical approaches have been conducted to evaluate them, including surveys of municipal planners (e.g., Clark et al., 2021), case studies (e.g., Parsons & Barling, 2022; Schleiffer et al., 2022; Cerutti et al., 2016; Morley & Morgan, 2021), simulated welfare analyses (e.g., Plakias, 2021), political economy analyses (e.g., Heenan et al., 2022), social network analyses (Gaitán-Cremaschi et al., 2022), and econometric models assessing, for example, the determinants of adoption of particular food procurement strategies (e.g., organic foods in Filippini et al., 2018) or of the probability of continuing FTS programming based on school and community characteristics (e.g., Bonanno & Mendis, 2021).

We argue that an additional, valuable perspective to food policy evaluation is to compare the economic benefits accruing to the full supply chain via the direct, indirect, and induced impacts generated by the policy. Benefits depend not only on what industries are affected directly by the policy, but to the degree that those industries transact with other local industries to support that direct activity expansion. We also contend that positive net economic impacts to the public policy authority (NYS in our application) are a necessary political condition to the longevity of such policies that support the multiple, nonmarket-valued objectives often advocated (e.g., rural development, family farms, food security, sustainability).

Incentives and policies affecting local food procurement by SFAs can be viewed as economic development strategies through import substitution and spillover effects that improve access for farmers and food manufacturers to new markets aiding in their growth and viability (Becot et al., 2017). However, empirical evaluations are often constrained - purchasing data are complex, often unorganized, and in varied forms from alternative vendors. To that end, most estimates of economic impacts from FTS programs analyze hypothetical or anticipated changes in economic activity (e.g., Holland et al., 2015; Tuck et al., 2010; Haynes, 2010). Others utilize aggregate food marketing data from SFAs to identify likely local suppliers (Gunter, 2011), consider aggregate farm sales data (Kluson, 2012), or assume a change in local institutional spending to estimate economic impacts (Roche et al., 2016).

Other attention has focused on the economic implications of FTS initiatives based on the propensity of local food producers to spend more on labor and other expenses per dollar of output that can lead to differences in job creation and spillover effects (Hughes & Boys, 2015; Shideler et al., 2018; Duval, et al., 2019), as well as specific benefits to farmers shifting to this market channel (Pinchot, 2014; Becot et al., 2017). When considering the policy mechanism of increased meal reimbursements to SFAs, Long et al. (2021) find that even an incentive of $0.05 per meal reimbursement can impact local school food purchasing decisions, although the economic impacts and implications of those shifts are not explored.

Research assessing economic impacts of FTS activities commonly use IO model software and data such as IMPLAN (Impact Analysis for PLANing), RIMS (Regional Input-Output Modeling System), or EMSI (Economic Modeling Specialist International), but with implementation that varies in level of detail, access to data, and model customization.[2] As mentioned above, early research explores economic impacts of hypothetical changes in farm production and pricing relevant to growing FTS sales. For example, Tuck et al (2010) look at potential impacts of F2S programs in Central Minnesota by developing different pricing scenarios and exploring ways in which farmers could receive alternative prices for their products. Similarly, Holland et al. (2015) explore potential economic impacts associated with a proposed FTS initiative in South Carolina. They model alternative scenarios of increases in crop production that would be expected under increased demand for local food products in schools, although noting that the aggregated crop production industry in their model includes all crops in the state, including those often not utilized by schools.

More recently, research has included actual school food purchase data but at more aggregate levels. For example, Duval et al. (2019)

[2] More information on IMPLAN, RIMS, and EMRI is available at: https://implan.com/, https://www.bea.gov/resources/methodologies/RIMSII-user-guide, and https://lightcast.io/, respectively.

evaluate the economic impacts of alternative scenarios involving opportunity costs, resource constraints, and export substitution informed by data from the Farm-to-School Census and aggregate school spending through government programs. Christensen et al. (2019) present a framework for evaluating the economic impacts of FTS programs by combining primary and secondary data to customize an IO model, reflecting the complex supply chains that link producers and schools, and apply it through two case studies representing Georgia and Minneapolis (MN) public schools. With data limited to total school food expenditures and the percentage of those total purchases expended locally, they estimate the number of farmers selling to the school and apply the totals through farm expenditure patterns curated from primary data collection.

Kluson (2012) obtain food sales data from farms in Sarasota County to assess changes in their institutional (school) sales, while Watson et al. (2018) obtain data on school food purchases from Sarasota County (Florida) schools to analyze purchasing patterns but did not estimate economic impacts of the local spending. As far as we are aware, there have not been any economic impact analyses that evaluate detailed food purchasing data by schools in the presence of a new local purchasing public incentive.

3. 30% NYS Initiative

Once a SFA satisfies the Initiative's local spending threshold, it receives an additional $0.191 in state reimbursement for each reimbursable lunch meal claimed in the subsequent school year. Reimbursement dollars must be maintained in the SFA's nonprofit food service account and be used in the same way as the other SFA budget funds. By rule, the Initiative is only based on the food served at lunch under the NSLP paid for with school budget dollars. Consequently, there is additional complexity involved in reporting by SFAs since they must distinguish the NYFPs that are served at lunch, even though breakfasts and other meals may use some of the same food products. For the purposes of this analysis, whether the SFA's spending specifically contributed to the Initiative threshold or not is technically irrelevant. We analyze all food procurement done by the SFA to see how the Initiative affected overall spending patterns. Indeed, the policy itself may influence spending within and across meal selections.

For the 2018/19 SY, seven SFAs were successful in qualifying for the higher reimbursement. Six of the SFAs had average daily participation (ADP), or average number of lunches served per day, below 1,900, BCSD's was 29,520. In total, BCSD represented 77% of total student enrollments, 85% of ADP, and 90% of food costs for lunch. For 2018/19, BCSD reported over $8.5 million in total food costs for lunch, of which $2.6 million was classified as NYFP purchases, or 31%. As reported by BCSD, they had to adapt their spending patterns to achieve the 30% threshold.[3]

Notably, all qualifying SFAs represent relatively low household income areas. SFAs with lower household incomes have higher proportions of students participating in lunch and with a higher percent qualifying for free- and reduced- cost lunches reimbursable through the NSLP. BCSD is the second largest school district in the state and one of the poorest, with more than 86% of its students eligible for free- or reduced-price lunch (USDA, 2021b). As such, the increased reimbursement under the Initiative is important for schools in lower income areas due to higher and costlier lunch demands. Indeed, 28% of the students in the qualifying SFAs lived at or below the poverty line, considerably higher than the state average of 18% (Bilinski et al., 2022). The average food cost per student across qualifying SFAs ranged from $157 to $291, the highest cost attributable to BCSD.

[3] These are nominal dollar values reported by BCSD and differ from our estimates as we calculate local differently through an IO framework. The differences are explored more in the data and methods sections that follow.

4. Data

We collect detailed food purchase data from BCSD for the 2017/18 and 2018/19 SYs. The data come from both government and private vendor sources, in varied formats, and some with handwritten notes crucial to identifying local versus non-local purchases. The data were standardized, coded as local or non-local, and margined appropriately (more on margining later).[4] Classification by intended meal was not requested given the research focus on total food procurement. The number of vendors utilized by BCSD to procure local food products increased in 2018/19 and some had to be contacted directly for access to more detailed purchasing data and local/non-local distinctions of products sold. A notable challenge in coding the data for the 2017/18 SY was that it was before the Initiative was in place and, therefore, BCSD did not categorize purchases in that year by local and non-local origin. In concert with BCSD, local purchases were defined retroactively by evaluating vendor and food product data and with direct contacts to vendors where necessary.

There are generally two sources of money SFAs use in purchasing food: federal entitlement dollars and local budget spending. Entitlement dollars are federal allocations provided to schools participating in the NSLP or other child nutrition programs through federal governmental offices and programs. Schools are allocated entitlement funds based on their ADP. NYS's Office of General Services (OGS) helps schools in the state obtain USDA food products by tracking their ADP and allocating entitlement dollars accordingly. Allocation of entitlement funds is an important component food procurement for schools; BCSD's represented 22% and 19% of total food purchases in 2017/18 and 2018/19, respectively (Table 1).

USDA Foods, the program through which entitlement allocations flow, is a program that supports SFAs by purchasing American grown and produced foods for use in schools. The types of foods available are limited to a list of about 30 products for which there is enough demand from SFAs. Entitlement dollars can also be used to divert USDA Foods products to processors for additional processing (e.g., whole chickens into chicken nuggets, apples into applesauce), but with restriction. It is the SFA's responsibility to utilize the entire product and only order quantities for which they can utilize at least 10% of the product per month to prevent waste. These large demand constraints mean that typically only very large processors (which tend to lie outside of NYS) can supply the needed quantities to USDA. As a result, general entitlement spending is an ineffective pathway for accessing local foods in NYS.[5]

The USDA Pilot Project for Unprocessed Fruits & Vegetables (USDA Pilot) emerged from the 2015 U.S. Farm Bill and allows SFAs in participating states to use existing USDA Foods entitlement allocations for additional vendors and designate a geographic preference (GP) to purchase fresh fruits and vegetables (USDA, 2022a). GP is available to schools in NYS (by law) in weighting factors beyond lowest price. NYS is one of eight states with access to the program to provide more flexibility for schools in their purchasing options.

The U.S. Department of Defense (DoD) Fresh Fruit and Vegetable Program (USDA DoD Fresh) also allows schools to use existing USDA Foods entitlement allocations to buy fresh produce (USDA, 2022b). USDA DoD Fresh operates through a partnership between the USDA and the DoD Defense Logistics Agency (DLA), as well as state distributing

[4] BCSD purchasing data were also available for the 2019/20 SY, however, drastic changes in school feeding programs due to the Covid-19 pandemic necessarily excluded it's use in our evaluation of the Initiative. However, 57 SFAs qualified for the Initiative in 2019/20 demonstrating program growth and benefit to SFAs.

[5] The process to become an approved USDA Foods supplier is rigorous; only 83 processors nationally are approved through USDA Foods (Econometrica, 2019, pp. 117-118).

Table 1
Food spending of Buffalo City School District (BCSD), by school year and source of funds (USD).

Source	2017–2018				2018–2019			
	NYS	Non-NYS	Total	% NYS	NYS	Non-NYS	Total	%NYS
Federal Entitlement:								
USDA Foods	14,971	2,681,789	2,696,760	0.6	22,999	3,174,620	3,197,619	0.7
USDA Pilot	96,997	102,315	199,312	48.7	0	0	0	–
Total	111,968	2,784,104	2,896,071	3.9	22,999	3,174,620	3,197,619	0.7
School Budget	4,646,691	5,404,348	10,051,038	46.2	5,688,273	8,160,543	13,848,816	41.1
Total Spending	4,758,658	8,188,451	12,947,109	36.7	5,711,272	11,335,162	17,046,434	33.5

Source: Buffalo City School District (BCSD) and authors' NYS/Non-NYS spending allocations. NYS = New York State, USA. Dollars are in nominal terms.

Table 2
Local food spending by Buffalo City School District, by industry and school year (USD).

Commodity	2018/19 ($)	2017/18 ($)	Change ($)	Change (%)
Fluid milk	2,538,805	2,308,427	230,378	10
Wholesale services (grocery & related products)	2,325,290	2,127,010	198,280	9
Fruit	245,788	154,472	91,316	59
Ice cream and frozen desserts	204,225	0	204,225	nd
Transportation services	160,760	128,628	32,132	25
Cheese	62,631	0	62,631	nd
Vegetables & melons	59,983	32,584	27,399	84
Canned fruits & vegetables	48,312	2,638	45,674	1,731
Frozen specialties	18,367	0	18,367	nd
Frozen fruits, juices, & vegetables	12,020	2,340	9,680	414
Beef from farms	11,250	559	10,691	1,913
Snack foods	10,550	0	10,550	nd
Meat processed from carcasses	5,940	387	5,553	1,435
All other food products	4,402	1,544	2,858	185
Grain from farms	2,628	0	2,628	nd
Other crops from farms	273	0	273	nd
Poultry & egg products	47	0	47	nd
Greenhouse, nursery, & floriculture products	0	71	−71	−100
Total	5,711,272	4,758,658	952,614	20

Source: Buffalo City School District and authors' NYS/Non-NYS spending allocations. Includes industries with positive local spending only. Full spending profiles available in Krasnoff (2022). Dollars are in nominal terms. nd = not defined.

agencies. DoD is available to any school in NYS under three geographical contract areas. It's focus on fresh produce provides added opportunities for local products.

Both USDA Pilot and USDA DoD Fresh are attractive to BCSD's general goal in securing more local products through their FTS programming. However, BCSD did not utilize USDA DoD Fresh in either 2017/18 or 2018/19, likely due to logistical constraints with DLA and defined contract areas, combined with difficulties in sourcing seasonal local produce for the menu year-round and with, oftentimes, higher costs. BCSD did utilize USDA Pilot in 2017/18 but not in 2018/19, when incentives changed dramatically for procuring local produce through budget spending to count for the Initiative. Utilizing USDA Foods entitlement allocations through these special programs (i.e., for produce) leaves less allocation for traditional USDA Foods entitlement purchases.[6] Even though the Initiative does not consider local purchases through entitlement allocations to count towards the 30% target, what is procured by SFAs with federal entitlement allocations is likely affected by implementation of the Initiative. For example, schools may choose to use entitlement dollars to purchase only those USDA Foods that cannot feasibly be purchased locally, while allocating more of their budget funds to NYFPs.

The second form of payment are dollars from school food budget spending, funded primarily through state and federal reimbursements, paid participation, and other sources of revenue (e.g., catering). Table 1 provides an overview of food procurement costs by BCSD in 2017/18 and 2018/19 (nominal dollars), by type of spending and author-computed local/non-local allocations.

Our estimated local spending percentages differ from those reported by BCSD for the Initiative, in part, since our data does not classify expenditures by intended meal. In qualifying for the Initiative, BCSD reported spending 31% of their school lunch budget (non-entitlement) on NYFPs in 2018/19 (Bilinski et al., 2022). Our estimate for local budget spending for all meals is 41%. As expected, the overall local percentage drops to 33% when entitlement spending is included. As reported by BCSD (Bilinski et al., 2022), total lunch spending for 2018/19 was $8.6 million, or about 50% of the total food costs shown in Table 1 ($17.0 million). A similar comparison is unavailable for 2017/2018 since local lunch spending was not tracked by BCSD before the Initiative was in place; based on our computations 46% of school budget spending was local in 2017/18.

An important reason why our estimates of local purchasing percentages differ from Initiative reporting relates to the way we assess and compute local dollars. As will be explained in detail below, we margin total costs for products into different industries. In different cases, this allows for more or less dollars to be allocated as local than the Initiative would count. The Initiative only considers whether products are NYFPs, it does not distinguish where portions of the total cost for wholesalers, producers, and/or transporters accrue. For example, if a NYFP is purchased from a non-local processor, the Initiative counts all dollars spent on this product as local. We, instead, margin out the processing fee as non-local and, thus, our count of local dollars on this product is lower. Alternatively, the Initiative does not allow for inclusion of any local

[6] While not shown, BCSD used both USDA Pilot and USDA DoD Fresh programs in 2019/20, but that was a consequence of pandemic conditions and a substantial increase in entitlement dollars available through the Summer Food Service Program for use during virtual, at home, school days.

dollars from purchases spent on non-local goods that are sold by a local wholesaler. We count the wholesale margin of these costs as local. As a significant portion of food is procured through wholesalers, this is a substantial difference.

To explore local spending shifts by food product category, Table 2 presents local nominal spending by commodity and year.[7] Aside from spending on fluid milk, wholesale services have the next largest portion of local spending. Margining is necessary for our estimation of changes in sales (and impact) to farm and food industries induced by the Initiative. In NYS, most schools provide local milk due to the state's prominent dairy industry. In addition to higher levels of local spending, the number of commodity groups increased in 2018/19 – from 11 to 17. Commodity categories added (in order of highest local sales) include ice cream and frozen desserts, cheese, frozen specialties, snack foods, grains, other crops, and poultry and egg products.[8] Increases in local commodity spending were also prominent for fruit, vegetables and melons, canned fruits and vegetables, and beef. The shifts in spending illuminate the primary ways in which BCSD's adapted to meet the 30% target.

5. Methodolgy

IO models distinguish the effects of a shock by the economic sectors of a geographically defined economy (NYS in our case). Since the effects of a shock do not distribute themselves evenly throughout the economy, IO methods estimate the extent of these impacts and trace how the changes impact different sectors of the economy. The analytical strength of this methodology is its ability to estimate indirect and induced economic effects stemming from the direct expenditures that lead to additional purchases by final users in an economy (Schmit & Boisvert, 2014). The direct effects are the initial set of expenditures applied to the IO multipliers that represent the change or the shock that results from a policy or project. The indirect effects are the additional business-to-business purchases that take place up the supply chain within the region stemming from the initial input (i.e., the direct effect). Induced effects are values of industry activity that stem from household spending of increased labor income that result from the initial input purchases and follow-on indirect effects.

Technically, our analysis utilizes IMPLAN's Social Accounting Matrix (SAM) model, rather than an IO model. A SAM incorporates not only economic data for an economy, but social data as well, including national and household income statistics (Van Wyk et al., 2015). Accordingly, the SAM has an input–output model at its core but has additional capacity to disaggregate households, firms, and other institutions such that the impacts and multipliers based on the SAM reflect ripple effects throughout the economy with somewhat greater precision than do those based on an IO model alone (Miller & Blair, 2009, Chapter 11).[9] Although the IO/SAM model is useful in assessing short-term economic impacts, it relies on assumptions including unlimited supply, constant prices, static framework, constant returns to scale, and fixed production technologies.

The IMPLAN SAM analysis provides the direct, indirect, and induced effects for employment, labor income, value added, and output. Employment represents average monthly jobs (both full and part time). Labor income sums the income earned by the employees and noncorporate proprietor owners. Total value added is comparable to gross domestic product (GDP) and includes labor income, taxes on production and imports, and other property type income. Output represents the value of annual industry production expressed in producer prices. For manufacturers output, is sales plus/minus changes in inventory. For service sectors, output equals sales. For retail and wholesale trade, it is the gross margin (not gross sales).

5.1. Vendor data processing

Efforts were made to obtain food purchase data at the most granular level to define local versus non-local expenditures across margined industries. Vendor data includes product name, cost, local or non-local distinction, and, in some cases, product formulation. Distinguishing which type of funds the products were purchased with (budget or entitlement dollars), allows us to estimate economic impacts separately.

Product purchases were mapped to appropriate IMPLAN industry codes. To facilitate this mapping, we utilize the NAICS/IMPLAN crosswalk (Clouse 2023).[10] The total cost of each item is divided into its margined components depending on type of vendor (farm, processor, or wholesaler) to properly allocate costs and local percentages. Raw (e.g., apples, eggs) and minimally processed foods (e.g., apple slices, chopped raw vegetables) were margined to applicable farm industry; processed products (e.g., cheese, lunch meat) were margined to applicable manufacturing industry. These two margins make up the "producer value" of the margin, the price received by the producer for the goods or services sold. Since most items purchased by BCSD were purchased through wholesalers, the total product cost is further margined through wholesale and transport margins.[11]

For most raw and minimally processed items, industry codes were easily mapped to the appropriate farm industry in IMPLAN (e.g., apples to fruit farming). For a relatively limited number of processed food products, the mapping to a particular NAICS or IMPLAN code was not clear and requiring author-judgement on the best fit.[12] For each type of product purchased from a wholesaler, IMPLAN commodity margins were applied to allocate the cost among wholesale, producer (farm or processor), and transport margins. Margins vary by commodity category.

Each margined component is assigned a local purchasing percentage (LPP) that indicates the portion of that cost component that occurs locally. For the wholesale margin, this was either 0% or 100% depending on if the wholesaler was located outside or inside NYS, respectively. For items purchased from a local (NYS) wholesaler but the item itself was not grown or produced in NYS, the wholesale margin

[7] The full spending patterns, local and nonlocal, are available in Krasnoff (2022, pp. 58-60).

[8] Per the NAICS-IMPLAN bridge categorization, ice cream and frozen desserts includes "ice cream, flavored ices, flavored sherbets, ice pops, and fruit pops" (Clouse 2023). For this category, BCSD spent most of the new local dollars on "grape slush" from Welch's in 2018/19.

[9] A typical SAM provides a mapping into a functional category for households usually based on household income class. The IMPLAN SAM serves this purpose with nine household income categories; however, with a shortcoming in the SAM accounts that restricts the full evaluation of income distribution effects (Alward & Lindall, 1996).

[10] The North American Industry Classification System (NAICS) is the standard used by the Federal statistical agencies to classify business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy. IMPLAN industries are based on NAICS codes but represent differing levels of NAICS code rollups. Generally speaking, manufacturing industries are at the 4–5 digit NAICS detail; whereas, agriculture and services are at the 3–4 digit NAICS detail.

[11] We use the wholesale margin for IMPLAN industry 398: Wholesale services – Grocery and related product wholesalers. For coding purposes, and practicality for most food purchases, the transport margin is restricted to the truck transportation services industry in IMPLAN (417). Specifically, once the wholesale and producer margins were allocated, the residual was assigned to IMPLAN industry 417.

[12] We used other databases, such as Reference USA and Data Axle that report NAICS codes for individual companies, to help facilitate our mapping process. Since the number of food products (and level of expenditures) not easily mapped was limited, there will be no discernible effect on the gross impacts estimated. Margining processes on specific food products decisions are detailed in Krasnoff (2022, pp. 61-63).

component is captured as local, but not the cost of goods sold. The producer margin LPP is based on the value of raw products (farm) or ingredients (processor) that were local; if no detailed information is provided, IMPLAN's default LPPs are used.

Where additional information about the local nature of product is known, industry production functions were edited to accommodate the additional detail. For example, adjusting the milk input LPP to 100% for cheese that is documented as using only NYS milk, or adjusting all LPP food inputs in the spending pattern to 57% for an eggroll documented as using 57% NYS ingredients. Some NYS processed food products sold by local processors were stipulated with a particular percentage of some local ingredients (e.g., processed hot dogs with 51% local beef). In this case, 100% of the processor margin is coded as local and the allocation of local spending among inputs within the production function adjusted accordingly.

The LPP of the transport margin is determined based on the origin of food product (farm or processor): 100% if local or IMPLAN's default LPP if non-local (55%). Intuitively, this makes sense since, even for non-local food products, some local transportation is needed for delivery.

5.2. Entitlement data processing

Food product purchases from entitlement spending are more complicated, requiring alternative coding for inclusion in the model. Items ordered from USDA Foods can only be traced to their most recent origin using State of Origin (SOO) reports posted by the FNS (USDA, 2020). The origin represents where the product was processed and packaged and may not fully represent where the products were grown or raised. SOO reports also list the value of products procured by state in the respective fiscal year. If NYS appeared in the SOO reporting for a commodity, the dollar value in NYS is divided by the total dollars spent on the commodity to estimate the LPP. Although schools cannot distinguish where their delivery of this commodity is coming from on the SOO list specifically, the computed percentage as the LPP for these items is a reasonable approximation. Notably, only eggs and apples through USDA Foods lists any contributions from NYS on the SOO reports.

We also collected USDA Processed Foods invoice data, which are related to the USDA Foods list when products from the latter are purchased and routed for further processing prior to delivery to BCSD. In this case, vendors are matched across the two lists and costs on each are additive entitlement purchases. In other words, USDA Foods costs represent the producer value and the USDA Processed Foods costs represent additional processing costs. Costs were coded separately to account for situations where the final product delivered was attributed to a different industry than the initial USDA Foods product (e.g., "raw whole chickens" diverted for further processing as "processed poultry products").

5.3. Model construction

For input into the customized NYS IO/SAM model, local spending was aggregated across vendors by year, industry code, and local/non-local spending allocation. Entitlement and budget spending (in nominal terms) were modeled separately each year. Since IMPLAN model years are based on calendar years (January through December) rather than SYs (July through June), we apply the ending calendar year in IMPLAN for each SY; i.e., 2017/18 and 2018/19 SY spending is applied to 2018 and 2019 model years, respectively, in IMPLAN.

Of primary focus is how BCSD's changes in local spending induced by the Initiative changed economic impact to the state economy. However, both the levels of spending and the underlying structural economy are changing year over year; i.e., both effect impact calculations. The complete change in impact to the NYS economy accounts for both changes. While it may be useful to consider the incidence of the change accruing from the two components, given that we consider only two consecutive years, we restrict our attention to changes in impact that encompass both components.[13] The change in spending by category is included in Table 2, however given customization within production functions for some individual commodity purchases across years, we model the local spending as direct effects each year and then compute the change in total impact across years. In other words, nominal spending levels are used each year with the corresponding (customized) model year construction in IMPLAN.

Importantly, we compare the change in gross economic impact from 2017/18 to 2018/19 to the negative impact of a tax effect on NYS households based on the cost of the additional state reimbursement. Put differently, if the Initiative is funded through an increase in the NYS budget by residents through income taxes (or other means), there is less income available to be spent by those taxpayers privately for other purposes. Alternatively, if funding for the Initiative represents a reallocation of previously budgeted state expenditures, the negative impact would follow from lower funding to the prior use. For clarity and ease of exposition, we chose the former.[14]

The added reimbursement cost is estimated by multiplying BCSD's ADP for 2018/19 (29,146 lunches) by the increased reimbursement through the Initiative ($0.191/meal) times the number of school days (180), or just over $1 million ($1,002,039). Although the reimbursement qualified for in one year is based on the ADP in the following year, we attribute the cost as a negative household income change in the qualifying year.[15] The actual amount BCSD received in 2019/20 was less than this amount as BCSD pivoted to emergency, at home food delivery due to the pandemic. Even so, we include the full potential tax effect to provide a more conservative measure of net economic impact.

6. Empirical results

The direct effects in the models represent local food spending by the BCSD each year; in so doing, they effectively represent the first round of indirect effects of food procurement through entitlement and budget funds. That spending in an IO/SAM framework stimulates additional indirect and induced effects to satisfy increased procurement demands. Changes in student enrollments (and ADP) over time will differentially affect food cost expenditures and reimbursements based on the number of students receiving free- or reduced-cost lunches; that effect is negligible in our results as enrollments increased less than 0.3% in 2018/19.[16]

6.1. Economic impact by source of funds

The gross economic impact results broken out by source of funds are presented in Table 3, in nominal terms. The results highlight the economic impacts of local food spending by BCSD each year with respect to employment, labor income, total value added, and output. Given the nature of the local products purchased through each funding source, it is expected that the overall multiplier effects (i.e., the total effect divided by the direct effect) are higher for budget spending than for entitlement spending. For example, dairy farm and dairy manufacturing industry

[13] In IMPLAN terms, we utilize "model year" 2018 for the 2017/18 SY and "model year" 2019 for the 2018/19 SY, prior to model customizations described in this section. Decomposition of the changes in spending and structural (model year) effects is available in Krasnoff (2022, pp. 47-49).

[14] Including only the additional reimbursements as the cost in our analysis increases the benefit cost ratio as it ignores additional induced industry effects. We prefer our approach as it represents a more conservative estimate of benefits relative to costs.

[15] IMPLAN has nine household income categories that vary in their spending patterns. For convenience, we apply the full negative shock to the middle-income household spending pattern of $30,000-$40,000 as an average effect.

[16] Total student enrollment was 38,243 in 2017/18 and 38,351 in 2018/19.

Table 3
NYS economic impacts of local food spending by the Buffalo City School District (BCSD), by school year and source of spending (USD).

Impact	Employment (jobs)	Labor Income ($)	Value Added ($)	Output ($)
2017–2018 Entitlement				
Direct	0.94	42,105	61,185	111,968
Indirect	0.28	21,161	30,637	49,048
Induced	0.27	17,361	30,524	46,834
Total	1.48	80,627	122,346	207,849
Multiplier	1.57	1.91	2.00	1.86
2017–2018 Budget				
Direct	16.8	1,204,813	1,649,925	4,646,691
Indirect	14.57	1,056,286	1,608,543	3,489,814
Induced	9.59	618,963	1,088,479	1,670,107
Total	40.96	2,880,063	4,346,948	9,806,612
Multiplier	2.44	2.39	2.63	2.11
2018–2019 Entitlement				
Direct	0.37	7,520	19,222	22,999
Indirect	0.01	801	1,250	2,206
Induced	0.03	2,288	4,099	6,235
Total	0.42	10,609	24,571	31,441
Multiplier	1.14	1.41	1.28	1.37
2018–2019 Budget				
Direct	22.78	1,480,883	2,068,552	5,688,273
Indirect	17.18	1,200,314	1,977,998	4,265,067
Induced	11.07	734,994	1,316,815	2,003,364
Total	51.03	3,416,191	5,363,366	11,956,704
Multiplier	2.24	2.31	2.59	2.10

Sources: Buffalo City School District (BCSD), authors' NYS/Non-NYS spending allocations, and author-customized NYS IMPLAN model. NYS = New York State. Dollars are in nominal terms.

products (primarily procured through budget spending) have larger relative multipliers in NYS than fruit and vegetable production and manufacturing (procured through entitlement allocations). The results will also vary depending on differences in margining, LPPs, and changes in production function parameters across products as discussed above.

The decrease in multipliers for entitlement expenditures between 2017/18 and 2018/19 is largely due to overall structural changes/inter-industry linkages (i.e., different model years in IMPLAN) in the economy that saw all farm industry output multipliers decrease somewhat, particularly for fruit and vegetable farming industries; i.e., from 1.70 to 1.30, and 1.79 to 1.53, respectively (Schmit 2021). The large decrease in total impact for entitlement spending is due to the shift away from USDA Pilot to budget spending. The decreases in multipliers for budget spending are moderated due to a relatively higher reliance on dairy products whose industry multipliers changed very little (i.e., the output multiplier for dairy farming changed from 1.98 to 1.89 and the fluid milk manufacturing multiplier was unchanged) and a mitigating effect from wholesale services, whose output multiplier increased slightly in 2018/19; i.e., from 1.94 to 1.98 (Schmit 2021).

6.2. Gross and net economic impacts

Gross economic impacts of BCSD's local spending due to the Initiative are summarized in the upper half of Table 4. Gross impacts are calculated as the economic impacts of local food spending in 2018/19 less the economic impacts of local food spending in 2017/18, or approximately nine jobs, $466,000 in labor income, $919,000 in value added, and $1,974,000 in industry output in nominal terms. Converting the results to constant year (2018/19) prices by the Producer Price Index (PPI) for food manufacturing (USBLS, 2023) reduces the net change in effects accordingly, albeit modest.

The estimated tax effect of the added $1,002,039 in lunch reimbursements is applied as a household income change in our IO model, with the results summarized in the lower half of Table 4. Given the tax represents a spending change out of household income, the results are all induced effects considering the initial $1 million shock and indirect and induced industry affects that stem from it. The negative impacts are approximately seven jobs, $450,000 in labor income, $839,000 in value added, and $1,280,000 in output.

Subtracting the impact effects of the tax from gross impact of the local spending changes reveals the net economic impacts of the Initiative to the state for BCSD's participation in it. Considering nominal dollars, net changes are modest but positive across all metrics: nearly 2 jobs, $7,000 in labor income, $80,000 in value added, and $700,000 in industry output. When converting to real terms the net change in labor income becomes modestly negative (-$12,243).

The driving goal of the Initiative is to support increases in local food products that originate from NYS farms. In this context, we present the distribution of net output effects by industry aggregate in Table 5. The net output effect of BCSD's increase in NYFP purchases is estimated to increase NYS farm sales by nearly $400,000, second only to the growth in NYS's food manufacturing industry sales of nearly $677,000, and above the increase in wholesale services ($239,431). The largest negatively affected industries result from major expenditure categories for households (i.e., health care, finance and insurance, housing, and retail purchases), albeit offset some from industry gains of increased local food purchases.[17]

7. Conclusions and policy implications

In the United States, local and regional food system policies have become abundant at both state and federal levels. Despite considerable differences across space, the priorities of these policies are often very similar; e.g., healthy food access, anti-hunger, dietary nutritional

[17] A detailed breakdown of all indirect and induced effects by industry and year are available in Krasnoff (2022, pp. 39-41).

Table 4
Gross and net NYS economic impacts of local food spending by the Buffalo City School District (BCSD) from the 30% NYS Initiative (USD).

Impact	Employment (jobs)	Labor Income ($)	Value Added ($)	Output ($)
2017–2018 - local food spending (budget + entitlement)				
Direct	17.74	1,246,917	1,711,110	4,758,658
Indirect	14.85	1,077,448	1,639,180	3,538,862
Induced	9.86	636,324	1,119,003	1,716,941
Total	42.45	2,960,690	4,469,294	10,014,461
Total ($2018/19)	42.45	2,979,822	4,498,175	10,079,175
2018–2019 – local food spending (budget + entitlement)				
Direct	23.15	1,488,403	2,087,775	5,711,272
Indirect	17.19	1,201,115	1,979,248	4,267,274
Induced	11.10	737,282	1,320,914	2,009,599
Total	51.45	3,426,800	5,387,937	11,988,145
Change in local food spending — 2018–2019 minus 2017–2018				
Direct change	5.41	241,486	376,665	952,614
Indirect change	2.34	123,667	340,068	728,412
Induced change	1.24	100,958	201,911	292,658
Total change	9.00	466,110	918,643	1,973,684
Total change ($2018/19)	9.00	446,978	899,762	1,908,970
2018–2019 increase in reimbursement, negative household income change				
Direct	0.00	0	0	0
Indirect	0.00	0	0	0
Induced	−7.10	−459,221	−838,766	−1,279,757
Total	−7.10	−459,221	−838,766	−1,279,757
Net economic impact 2018–2019				
Direct change	5.41	241,486	376,665	952,614
Indirect change	2.34	123,667	340,068	728,412
Induced change	−5.86	−358,263	−636,855	−987,099
Total change	1.90	6,889	79,877	693,927
Total change ($2018/19)	1.90	−12,243	50,996	629,213

Sources: Buffalo City School District (BCSD), authors' NYS/Non-NYS spending allocations, and author-customized NYS IMPLAN model. NYS = New York State. Unless otherwise indicated, dollars are in current nominal terms. Real dollar conversion to $2018/19 uses the Producer Price Index (PPI) for food manufacturing (USBLS, 2023).

Table 5
Estimated gross and net output changes in NYS from Buffalo City School District's (BCSD) participation in the 30% NYS Initiative, by industry aggregates (USD).

Industry	Gross change	Tax effect	Net change
Total	1,973,684	−1,279,757	693,927
Food manufacturing	689,682	−12,997	676,685
Agricultural production	401,711	−1,916	399,795
Wholesale trade	299,974	−60,543	239,431
Management, scientific, & technical services	118,280	−79,089	39,191
Transportation & warehousing	70,380	−32,213	38,167
Forestry, commercial logging, fishing, & hunting	165	−78	87
Mining & drilling	−71	−432-	−503
Non-Ag manufacturing	13,085	−15,941	−2,856
Administrative & waste services	30,299	−33,825	−3,527
Agricultural and forestry support services	−4,149	−58	−4,207
Construction	1,499	−9,472	−7,973
Government	6,609	−19,495	−12,886
Arts, entertainment, & recreation	5,981	−20,381	−14,400
Educational services	4,889	−19,459	−14,571
Utilities - generation & distribution	8,749	−28,549	−19,800
Other services	20,247	−56,677	−36,430
Accommodations & food services	23,461	−60,675	−37,215
Information & communications	36,439	−76,297	−39,859
Retail trade	26,075	−107,156	−81,081
Real estate & rental	112,821	−236,929	−124,108
Finance & insurance	62,053	−192,664	−130,611
Health & social services	45,505	−214,910	−169,404

Sources: Buffalo City School District (BCSD), authors' NYS/Non-NYS spending allocations, and author-customized NYS IMPLAN model. Dollars are in nominal terms. Note: Gross output changes include direct, indirect, and induced effects that are additive across sectors. NYS = New York State. Industry aggregate effects are arranged in descending order by the net change.

improvements, local food production, and sustainable food systems (Bassarab et al., 2019). Public policy proponents and academics have used a variety of conceptual frameworks and empirical methods to assess the proficiency and efficacy of food policies in meeting stated objectives.

Through a quasi-experimental approach, we analyze a new FTS state-level policy aimed at increasing public procurement of local foods in NYS. We adapt and extend an IO-based framework that allows for margining of industry expenditures and meticulous customization of food industry production functions and local purchasing coefficients using granular food purchasing data. Although previous IO research in this space has importantly shed light on the ways that different types of spending can produce different levels of impact, the lack of assessment of actual, detailed purchasing data leaves ambiguity as to what the actual impacts may be. A customized IO approach combined with detailed purchasing data over time provide support to our empirical results beyond a 'simple' impact study. Our approach is unique in that we analyze the additive impact of a change in spending that results from a specific policy initiative.

Of primary focus is whether the costs of the Initiative are more or less than the economic impacts achieved from it. As a result of the Initiative, we observe clear shifts in food spending categories that suggest changes in what and where foods were purchased. Considering changes in gross state product (GSP or value added) as an evaluative measure, policy-induced changes in local food procurement by the BCSD increase contributions to GSP by nearly $900,000 ($899,762) in constant year (2018/19) prices. Public subsidies (increased meal reimbursements) to support those procurement changes imply a reduction in GSP of nearly $839,000 ($838,766). Taken together, benefits exceed costs, or a benefit cost ratio of 1.06 (899,762/838,766). Put differently, for every dollar of value added lost in the state to support the program, $1.06 of value added are added. Importantly, SFAs have been known to stop tracking their local purchases once they cross the 30% threshold. Ensuring that SFAs track local and non-local spending through the entirety of the SY will provide a more accurate representation of impact. Changing qualification requirements to include full-year reporting is a simple and reasonable fix to this issue.

The decomposition of total industry impacts (Table 5) highlights projected industry 'winners' and 'losers' (as we define by net changes in industrial output) as a result of the policy. The reimbursements funded by taxpayers (lower household incomes) most negatively affect industries representative of their primary budget expenditures and may exceed the benefits to those industries from the farm and food industry expansion. The net gain overall and the strong gains in agricultural sectors specifically align well with enduring political support and overall policy goals, respectively.

Lunch spending is a significant portion of total food procurement for SFAs but ignoring other meal spending in the policy underestimates local impact and ignores reallocation effects. While our estimate of the overall percentage of local purchasing dollars decreased in 2018/19 (relative to the stated increase by BCSD), the total dollars of local spending increased by over $950 thousand ($952,614) in nominal terms – spending changes, not percentage changes, drives impact. Our differences suggest shifting within the total food spending budget occurred for local foods and for what meal. Including local spending on other meals as part of the Initiative will reduce administrative burdens of SFAs and increase incentives to expand local purchases.

There is no direct coordination of or consideration for local food spending beyond budget funds for the Initiative. Implementing new local procurement policies without considering the impact on preexisting policies may result in a combined impact that is lower than expected; i.e., the sum of impacts in isolation is greater than the impact of programs operating collectively. While traditional entitlement dollars through USDA Foods provide little opportunities for local procurement, more recent developments of federal programs that focus on fresh produce increases opportunities for local procurement based on perishability considerations alone. If programs also allow for geographic preference as part of food procurement decisions, local procurement opportunities expand even further. While implementing local accounting rules for SFAs consistent with our customized IO framework is impractical, state-level policy development must consider potential unintended consequences on related policy participation to truly maximize benefits.

The limitation of our approach to one school district and two years of data is important in the context of the robustness of our results relative to the nature of the policy mechanisms through which it operates. The quadrupling of state reimbursement rates – reimbursements that are a function of the number of free- and reduced-price lunches served – clearly incentivize SFAs in lower household income areas to participate. In other words, directing the subsidies through this mechanism inherently guides policy beneficiaries to schools and communities in low-income areas. Because of this specific focus, the robustness of our results aligns well with similar FTS food policies directing attention to low-income areas – something common in FTS policies advocating for improved food access and equity issues. SFAs are guided in their food procurement and meal preparation decisions on mandatory nutritional constraints for students. These constraints help expand the robustness of our results beyond a single school district.

A comparison over a limited time frame limits the impacts of other structural changes or acute events in the economy from clouding specific policy impacts, however, what one chooses for the baseline is critical in these (before and after) types of analyses. That said, utilizing food procurement data in the year preceding and the year of the policy intervention are arguably the most accurate representation of the structural economy. Furthermore, it is the shift in local purchasing incident to the policy change that is of most importance. In this sense, the IO results represent an expected annual change in impact to the degree that SFAs maintain their local purchasing going forward, something necessary to maintain a minimum of 30% local purchases.

An important limitation of our IO approach (as with all IO analyses) is that the direct effects modeled (the change in local food spending in our case) are assumed to be new deliveries to (an increase in) final demand; i.e., the value of goods and services produced and sold to final users. Clearly, the demand for local food products by BCSD because of the policy have increased, but does the amount of that change continue to hold when considering the final demand by all users of those products in the state? If the increase in local purchases by BCSD is, instead, a reallocation of products previously sold by suppliers to other local buyers, there is no change in final demand, except perhaps to price effects. Without knowing actual producer changes in response to the increase in local food procurement demand, the net benefit of the Initiative at BCSD can be considered an upward bound. However, as an approximation, at least 67% of the increase in local spending by BCSD must represent new sales to final demand (i.e., $1.28 M tax output impact divided by $1.91 M food spending output impact (Table 4) to retain a benefit cost ratio above unity.

To provide context for this result, we return to the USDA Local Food Marketing Practices Survey mentioned in the introduction section that noted a large percentage increase in direct farm sales to institutions and intermediaries (e.g., educational and medical facilities). Between 2015 and 2020, direct sales to institutions and intermediaries in the Northeast region increased over $462 million, while direct sales to retail (e.g., supermarkets, restaurants, caterers) increased nearly $190 million; however, direct sales to consumers (e.g., farmers markets, farms stands, online marketplaces) actually decreased by $115 million (USDA, 2022c).[18] If all of the decrease in consumer sales is a reallocation to these alternative markets (a strong assumption), this represents an 18% reallocation effect (i.e., the drop in consumer sales relative to the gains in other markets). Put differently, 82% of the positive change in sales in

[18] NYS data were not available.

these other markets can be interpreted as an increase in final demand. Some reallocation of local market sales is expected, however, the 67% threshold in our sensitivity analysis appears reasonable in the context of strong overall market changes.

The Initiative's driving goal is to increase expenditures on NYFPs through taxpayer financed subsidies, clearly recognizing the benefit of ripple (multiplier) effects of local spending. Evaluating the extent of that growth via the directly and indirectly affected industries is one, but not the only, means to that end. We recognize that our focus on benefits is limited to defined market-based economic effects that either directly or indirectly ignores other values-based measures and nonmarket valued externalities. Our intent is not to dismiss their importance and, to the degree our definition of benefit exceeds the costs, provides necessary political support for such policies to continue.

More understanding of the shifts in spending and benefits can occur through inclusion of data for additional years and SFAs, conditional on other broader market-related conditions (e.g., the pandemic) and SFA/geographic area characteristics (e.g., agricultural and manufacturing activity, socio-economic demographics) are appropriately accounted for. Doing so may require alternative metrics of benefit and empirical (e.g., econometric) approaches. Changes over time in policy design may also present opportunities to consider treated and control groups relative to our before and after approach. Nonetheless, our results have important implications and insights for FTS policy makers and SFAs surrounding local food purchasing incentives. Our efforts also provide a needed pathway and replicable framework for other SFAs and interested parties to begin analyzing their projected impacts irrespective of location and specific FTS policy/program in question.

CRediT authorship contribution statement

Shayna M. Krasnoff: Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing. **Todd M. Schmit:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Cheryl B. Bilinski:** Formal analysis, Investigation, Resources, Data curation, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We wish to thank the Buffalo City School District and their food vendors for access to detailed food procurement data and repeated interactions during the data collection and management processes. Personnel to support this research was made possible through a USDA Agricultural Marketing Service Local Foods Promotion Program grant (AM190100XXXXG151) and a NYS Department of Agriculture and Markets grant. The funders played no role in the study design or the decision to submit the article for publication (CM04068FC). The views expressed are the authors' and do not necessarily represent the policies or views of any sponsoring agencies.

References

Alward, G., Lindall, S., 1996. Deriving SAM multipliers using IMPLAN. Paper presented at the IMPLAN Users Conference, Minneapolis, MN.

Aubry, C., Kebir, L., 2013. Shortening food supply chains: A means for maintaining agriculture close to urban areas? The case of the French metropolitan area of Paris. Food Policy 41, 85–93. https://doi.org/10.1016/j.foodpol.2013.04.006.

Bassarab, K., Santo, R., Palmer, A., 2019. Food policy council report 2018. Johns Hopkins Center for a Livable Future. https://assets.jhsph.edu/clf/mod_clfResource/doc/FPC%20Report%202018-FINAL-4-1-19.pdf.

Bauman, T., Jablonski, B.B.R., 2018. The financial performance implications of differential marketing strategies: Exploring farms that pursue local markets as a core competitive advantage. Agricultural and Resource Economics Review 47 (3), 477–504. https://doi.org/10.1017/age.2017.34.

Becot, F., Kolodinsky, J., Roche, E., Zipparo, A., Berlin, L., Buckwalter, E., McLaughlin, J., 2017. Do farm-to-school programs create local economic impacts? Choices. 32, 1–8. https://www.choicesmagazine.org/UserFiles/file/cmsarticle_565.pdf.

Bilinski, C., C. Bull, & B. O'Conner, B. 2022. 30% NY Initiative: Opportunities, barriers, and pathways to success. HarvestNY, Cornell Cooperative Extension. https://harvestny.cce.cornell.edu/uploads/doc_217.pdf.

Bonanno, A., Mendis, S.S., 2021. Too cool for farm to school? Analyzing the determinants of farm to school programming continuation. Food Policy 102, 102045. https://doi.org/10.1016/j.foodpol.2021.102045.

Brekken, C.A., Dickson, C., Peterson, H.H., Feenstra, G., Ostrom, M., Tanaka, K., Engelskirchen, G., 2019. Economic impact of values-based supply chain participation on small and midsize produce farms. J. Food Distrib. Res. 50 (2) https://doi.org/10.22004/ag.econ.300074.

Cerutti, A.K., Contu, S., Ardente, F., Donno, D., Beccaro, G.L., 2016. Carbon footprint in green public procurement: Policy evaluation from a case study in the food sector. Food Policy 58, 82–93. https://doi.org/10.1016/j.foodpol.2015.12.001.

Christensen, L., Jablonski, B.B.R., Stephens, L., Joshi, A., 2019. Evaluating the economic impacts of farm-to-school procurement: An approach for primary and secondary financial data collection of producers selling to schools. J. Agric. Food Sys. Community Dev. 8 (3), 73–94. https://doi.org/10.5304/jafscd.2019.08C.002.

Clark, J.K., Conley, B., Raja, S., 2021. Essential, fragile, and invisible food infrastructure: The role of urban governments in the United States. Food Policy 103, 102014. https://doi.org/10.1016/j.foodpol.2020.102014.

Clouse, C., 2023. IMPLAN industries and NAICS correspondences. IMPLAN Support, IMPLAN, Huntersville, NC.

Cohen, N., Ilieva, R.T., 2021. Expanding the boundaries of food policy: The turn to equity in New York City. Food Policy 103, 102012. https://doi.org/10.1016/j.foodpol.2020.102012.

DeSchutter, O., Jacobs, N., Clément, C., 2020. A 'Common Food Policy' for Europe: How governance reforms can spark a shift to healthy diets and sustainable food systems. Food Policy 96, 101849. https://doi.org/10.1016/j.foodpol.2020.101849.

Duval, D., Bickel, A.K., Frisvold, G.B., 2019. Farm-to-school programs' local foods activity in southern Arizona. J. Agric. Food Sys. Community Dev. 8 (3), 53–72. https://doi.org/10.5304/jafscd.2018.08c.001.

Econometrica, Inc. 2019. USDA Foods trend analysis (SY15-18). Final Report submitted to USDA Food and Nutrition Service (Contract No. AG-3198-C-15-0019). Bethesda, MD (USA). www.EconometricaInc.com.

Filippini, R., De Noni, I., Corsi, S., Spigarolo, R., Bocchi, S., S., 2018. Sustainable school food procurement: What factors do affect the introduction and increase of organic food? Food Policy 76, 109–119. https://doi.org/10.1016/j.foodpol.2018.03.011.

Gaitán-Cremaschi, D., Klerkx, L., Aguilar-Gallegos, N., Duncan, J., Pizzolón, A., Dogliotti, S., Rossing, W.A.H., 2022. Public food procurement from family farming: A food system and social network perspective. Food Policy 111, 102325. https://doi.org/10.1016/j.foodpol.2022.102325.

Graham, H., Zidenberg-Cherr, S., 2005. California teachers perceive school gardens as an effective nutritional tool to promote healthful eating habits. J. Am. Diet. Assoc. 105 (11), 1797–1800. https://doi.org/10.1016/j.jada.2005.08.034.

Gunter, A. 2011. Rebuilding local food systems: Marketing and economic implications for communities. Master's Thesis. Colorado State University. https://dspace.library.colostate.edu/bitstream/handle/10217/49814/Gunter_colostate_0053N_10765.pdf.

Haynes, M. 2010. Farm-to-School in Central Minnesota: Applied economic analysis. Community Assistantship Program, University of Minnesota. https://hdl.handle.net/11299/195481.

Heenan, M.S., Jan, K.C., Shanthosh, J., 2022. A political economy analysis protocol: Case study implementing nutrition and sustainability policy into government food procurement. PLoS One 17 (9), e0274246. https://doi.org/10.1371/journal.pone.0274246.

Holland, J.H., Thompson, O.M., Godwin, H.H., Pavlovich, N.M., Stewart, K.B., 2015. Farm-to-school programming in South Carolina: An economic impact projection analysis. Journal of Hunger & Environmental Nutrition 10 (4), 526–538. https://doi.org/10.1080/19320248.2014.980045.

Hughes, D.W., Boys, K.A., 2015. What we know and don't know about the economic development benefits of local food systems. Choices 30 (1), 1–6. https://www.choicesmagazine.org/UserFiles/file/cmsarticle_413.pdf.

Joshi, A., Azuma, A.M., Feenstra, G., 2008. Do farm-to-school programs make a difference? Findings and future research needs. Journal of Hunger & Environmental Nutrition 3 (2), 229–246. https://doi.org/10.1080/19320240802244025.

Khalife, G. 2018. In New York State, no student goes hungry. Food Policy Snapshot, Hunter College New York City Food Policy Center. https://www.nycfoodpolicy.org/in-new-york-state-no-student-goes-hungry/.

Kluson, R.A., 2012. Regional and local economic impacts of the Sarasota County farm to school program. University of Florida Extension, Fact Sheet. https://sfyl.ifas.ufl.edu/media/sfylifasufledu/sarasota/documents/pdf/ag/SarasotaCounty_FarmtoSchoolEconomicImpact.pdf.

Kovacs, V.A., Messing, S., Sandu, P., Nardone, P., Pizzi, E., Hassapidou, M., Brukalo, K., Tecklenburg, E., Abu-Omar, K., 2020. Improving the food environment in kindergartens and schools: An overview of policies and policy opportunities in Europe. Food Policy 96, 101848. https://doi.org/10.1016/j.foodpol.2020.101848.

Krasnoff, S.M., 2022. Economic assessment of farm-to-school food purchasing incentives: The case of the Buffalo City School District. Cornell University, Ithaca, NY (USA). Master's Thesis.
Long, A.B., Jablonski, B.B.R., Costanigro, M., Frasier, W.M., 2021. The impact of state farm to school procurement incentives on school purchasing decisions. J. Sch. Health 91, 418–427. https://doi.org/10.1111/josh.13013.
Martinez, S.W., 2016. Policies supporting local food in the United States. Agriculture 6 (3), 43. https://doi.org/10.3390/agriculture6030043.
Miller, R., Blair, P., 2009. Input -output analysis: Foundations and extensions, 2nd edition. Cambridge University Press, Cambridge, England.
Morley, A., Morgan, K., 2021. Municipal foodscapes: Urban food policy and the new municipalism. Food Policy 103, 102069. https://doi.org/10.1016/j.foodpol.2021.102069.
New York State Department of Agriculture and Markets (NYSDAM). 2022. Farm-to-School: Overview. https://agriculture.ny.gov/farming/farm-school.
Ozer, E.J., 2007. The effects of school gardens on students and schools: conceptualization and considerations for maximizing healthy development. Health Educ. Behav. 34 (6), 846–863. https://doi.org/10.1177/2F1090198106289002.
Parsons, K., Barling, D., 2022. Identifying the Policy Instrument Interactions to Enable the Public Procurement of Sustainable Food. Agriculture 12, 506. https://doi.org/10.3390/agriculture12040506.
Pinchot, A., 2014. The economics of local food systems: A literature review of the production, distribution, and consumption of local food. Extension Center for Community Vitality, University of Minnesota Extension. https://hdl.handle.net/11299/171637.
Plakias, Z.T., 2021. Cost-benefit analysis as a tool for measuring economic impacts of local food systems: Case study of an institutional sourcing change. J. Agric. Food Sys. Community Dev. 10 (3), 161–185. https://doi.org/10.5304/jafscd.2021.103.011.
Roche, E., Becot, F., Kolodinsky, J., Conner, D., 2016. Economic contribution and potential impact of local food purchases made by Vermont schools. Center for Rural Studies, University of Vermont. https://agriculture.vermont.gov/sites/agriculture/files/documents/Farm_to_School_Institution/Economic%20Contribution%20of%20Farm%20to%20School%20in%20Vermont%20.pdf.
Schleiffer, M., Landert, J., Moschitz, H., 2022. Assessing public organic food procurement: the case of Zurich (CH). Org. Agric. 12, 461–474. https://doi.org/10.1007/s13165-022-00402-5.
Schmit, T.M. 2021. The economic contributions of agriculture to the New York State economy: 2019. EB 2021-04, Charles H. Dyson School of Applied Economics & Management, Cornell University. https://dyson.cornell.edu/wp-content/uploads/sites/5/2021/09/EB2021-04_TShmit-VD.pdf.
Schmit, T.M. & R.N. Boisvert. 2014. Agriculture-based economic development in New York State: Assessing the inter-industry linkages in the agricultural and food System, EB 2014-03. Charles H. Dyson School of Applied Economics and Management, Cornell University, Ithaca, NY (USA). Department of Applied Economics and Management, http://publications.dyson.cornell.edu/outreach/extensionpdf/2014/Cornell-Dyson-eb1403.pdf.
Schmit, T.M., Jablonski, B.B.R., Bonanno, A., Johnson, T.G., 2021. Measuring stocks of community wealth & their association with food systems efforts in rural & urban places. Food Policy 102, 102119. https://doi.org/10.1016/j.foodpol.2021.102119.
Shideler, D., Bauman, A., Thilmany, D., Jablonski, B.B.R., 2018. Putting local food dollars to work: the economic benefits of local food dollars to workers, farms and communities. Choices 33 (3), 1–8. https://www.choicesmagazine.org/UserFiles/file/cmsarticle_646.pdf.
Tuck, B., Haynes, M., King, R., Pesch, R., 2010. The economic impact of farm-to-school lunch programs: A central Minnesota example. University of Minnesota Extension. https://hdl.handle.net/11299/171560.
United States Bureau of Labor Statistics (USBLS). 2023. Producer Price Indexes – PPI Databases (food manufacturing, 311). https://www.bls.gov/ppi/databases/.
United States Department of Agriculture (USDA). 2020. State of origin for USDA Foods. Food and Nutrition Service. https://www.fns.usda.gov/usda-foods/state-origin-usda-foods.
United States Department of Agriculture (USDA). 2021b. School Food Authority (SFA) Profile for Buffalo City SD, NY 14202. Farm to School Census. Food and Nutrition Service. https://farmtoschoolcensus.fns.usda.gov/census-results/states/ny/buffalo-city-sd-14202.
United States Department of Agriculture (USDA). 2021a. 2019 Farm to School Census Report (Summary). Food and Nutrition Service. https://fns-prod.azureedge.us/sites/default/files/resource-files/2019-Farm-to-School-Census-Summary.pdf.
United States Department of Agriculture (USDA). 2022c. 2020 Local Food Marketing Practices Survey (2015 and 2020 survey years, Northeast region). National Agricultural Statistics Service. https://www.nass.usda.gov/Publications/AgCensus/2017/Online_Resources/Local_Food/index.php.
United States Department of Agriculture (USDA). 2022a. Pilot Project: Unprocessed Fruits & Vegetables. Agricultural Marketing Service. https://www.ams.usda.gov/selling-food/pilot-project.
United States Department of Agriculture (USDA). 2022b. USDA DOD Fresh Fruit and Vegetable Program. Food and Nutrition Service. https://www.fns.usda.gov/usda-foods/usda-dod-fresh-fruit-and-vegetable-program.
United States Department of Agriculture (USDA). 2023. Integrating local foods into child nutrition programs. Food and Nutrition Service. https://www.fns.usda.gov/f2s/integrating-local-foods-child-nutrition-programs.
Van Wyk, L., Saayman, M., Rossouw, R., Saayman, A., 2015. Regional economic impacts of events: A comparison of methods. S. Afr. J. Econ. Manag. Sci. 18 (2), 155–176. https://doi.org/10.17159/2222-3436/2015/v18n2a2.
Watson, J., Treadwell, D., Bucklin, R., 2018. Economic analysis of local food procurement in southwest florida's farm to school programs. J. Agric. Food Sys. Community Dev. 8 (3), 61–84. https://doi.org/10.5304/jafscd.2018.083.011.